

①

Media company video streaming platform

cloud architecture:

The cloud architecture should include a Load Balancer, Auto Scaling Group, Application servers, content delivery network (CDN), cache, and object storage. The CDN delivers videos from the nearest server, while the cache stores frequently accessed content to reduce response time.

Load Balancing:

Load balancing distributes incoming user requests across multiple servers. It prevents server overload, improves response time, and ensures uninterrupted video streaming even during heavy traffic.

Auto Scaling:

Auto scaling automatically increases or decreases the number of servers based on user demand. During peak hours, additional servers are added, and during low traffic, extra servers are removed to reduce costs.

Cost optimization:

- Use CDN to reduce bandwidth usage.
- Store videos in low-cost object storage.
- Enable Auto scaling to pay for idle resources.
- Continuously monitor resource usage.

Conclusion:

By combining load balancing, auto scaling, caching and CDN, the platform can provide fast, reliable, and cost-effective video streaming.

② FinTech startup using containers

proposed solution:

The startup should use Docker containers managed by Kubernetes and deploy applications across multiple cloud providers.

Container orchestration:

Kubernetes automates container deployment, scaling, load balancing and recovery. It ensures that applications run consistently across different cloud platforms.

service Discovery and Scaling

Kubernetes provides service discovery through DNS and automatically scales containers based on CPU or memory usage, ensuring efficient resource utilization.

Benefits

- High availability
- No vendor lock-in
- Easy scalability
- Better disaster recovery

Risks:

- complex management
- Higher operational costs
- security and networking challenges

Conclusion

Kubernetes with a multi-cloud strategy provides reliable, scalable and consistent application deployments.

② Backup and Disaster Recovery

solution

The organization should use automated cloud backups, cross-region replication, snapshots, and object versioning to protect data from accidental deletion.

RTO and RPO

- Recovery Time objective: The maximum acceptable time to restore services after a failure
- Recovery point objective: The maximum amount of data that can be lost during a disaster.

Business continuity

- schedule regular backups
- Replicate data to another cloud region
- perform periodic disaster recovery testing.
- use automatic failover during failures.

conclusion:

Backup, replication, and versioning ensure quick recovery, minimal data loss, and continuous availability of the e-learning platform.

④ Hybrid cloud for Government Agency

Hybrid Architecture

Sensitive data should remain on-premises while non-sensitive applications can be hosted in the public cloud. A secure VPN or dedicated connection should be used for communication between both environments.

Identity management and Encryption

Implement a single sign-on, multi Factor Authentication, and Role-Based Access control. Encrypt stored data using AES-256 and data in transit using TLS/SSL.

Challenges

- complex infrastructure management
- data synchronization issues
- security and compliance requirements
- Higher maintenance costs.

conclusion:

A hybrid cloud model offers flexibility while ensuring the security and compliance of sensitive government data.

⑤ SaaS provider High availability

Multi-Region Deployment:

Deploy the application across multiple cloud regions with a global balancer. Replicate databases between regions to ensure data availability.

Data Replication and Failover

Data should be continuously replicated to the secondary region. If the primary region fails, traffic is automatically redirected to the backup region, ensuring uninterrupted service.

Monitoring and Alerting

Use monitoring tools to track CPU usage, memory, response time and application health. Configure alerts through email or SMS to quickly detect and resolve issues.

conclusion:

Multi-region deployment, data replication, automatic failover, and continuous monitoring provide high availability and fault tolerance for SaaS applications.